julia racer — iRacing test plan

IRPLAN-4, 19 October 2026 — what julia still needs from iRacing's Lotus 49 after the 18 October session

What the 18 October session settled

Thank you for the 18 October session: 38 telemetry files, 31 setup sheets and 3 AI races. What it settled in julia:

- **Anti-roll bars:** the five skidpad runs with front/rear arms soft and firm gave each bar its own stiffness law. julia's body roll per g now matches iRacing's to within 0.013°/g on all five.
- **Engine power and air:** your pulls showed iRacing's engine loses power in thinner air. julia now gives each track iRacing's air (Monza 1.151 kg/m³, Nürburgring 1.095–1.099), and its straight-line pulls went from 92 % to 98 % of yours.
- **Camber and the garage:** the setup sheets fitted how camber changes the corner weights and ride heights, so julia's Garage tab now shows the same numbers as iRacing's for every camber.

What it left open, and why this plan exists:

1. **Your accelerator only reached 80.7 %** in iRacing on 18 October (the files record it). Full-throttle tests at 80.7 % are part-throttle tests. That probably also explains the last 6 % of power missing at 8,000–9,000 rpm. **Part A fixes it first.**
2. **Part-throttle power.** julia's engine map at 25–75 % pedal was fitted before the air-density change and now disagrees with your files.
3. **Trail braking at 250 km/h** came out inconclusive: julia's car slid 5.0°, iRacing's 6.8°, on your inputs, but both runs were gentle. A firmer version settles it.
4. **The rear's grip under power** (the understeer item): julia's car needs 72 % throttle to hold a slide iRacing holds at 48 %. Steps of throttle on a steady circle measure exactly that.
5. **Car-to-car contact** (your new backlog item): your three AI races have collisions in them. A few deliberate, controlled contacts tell julia what a touch of wheels does.

6. **Two garage questions:** the “tire pressure max” sheet reads 179 kPa, but 207 kPa appears in an earlier skidpad file; and the bars’ stiffness for the holes you didn’t test (3, 4, 6) and for the other diameters is still estimated.

How to use this plan

Seven parts, about **2 hours 35 minutes** in all. Any part can be done on its own, in any order, **except Part A, which goes first.** Part B is in the garage; the rest are on track. Part G is driven twice, in iRacing and then in julia.

Part	What	Time	What it settles in julia
A	Pedal check (and recalibrate if needed) + full-throttle pulls	15 min	Full power at 8,000–9,000 rpm (ENGINERHO-1)
B	Garage: pressure range, list knobs, bar stiffness per hole	20 min	The pressure question; every bar hole (RARBARMS-1)
C	Part-throttle steps on the straight	15 min	The engine map at part throttle (ENGINERHO-1 S2)
D	Skidpad: throttle and brake steps on a steady circle	25 min	The rear’s grip under power and braking (UNDERSTEER-1)
E	Firm trail braking at 250 km/h	15 min	Braking stability at speed (TRAILBRAKE-DRIFT)
F	Controlled car-to-car contacts	25 min	How contact spins and launches cars (CARCOLLIDE-2)
G	WW103 side by side: the same tests in iRacing	40 min	Whether julia’s WW103 behaves as iRacing’s

Part	What	Time	What it settles in julia
	and in julia		(PHYSFID-1)

Car: Lotus 49. **Track:** Monza GP *without* chicanes, as before, unless a test says Skidpad. **Base setup:** 261010_rung_official (the default) unless a test says WW103 (261005_jr_ww103).

Before you start (2 minutes)

1. **Telemetry on** (Alt+L) for every on-track test, one file per test if you can (stop and restart telemetry between tests).
2. **Setup sheets, not screenshots,** for everything in the garage except the drop-down lists (B2). Export each sheet as on 18 October, **under the name this plan gives it** (the file name, and the setup name if iRacing asks you to save first). **Wait 3 seconds after each click** before exporting, until the numbers stop changing and no label has an asterisk.
3. **One knob at a time, from the base setup.** Reload the base before each new knob.
4. **Filing:** put everything on the USB stick under 261019/telemetry, 261019/sheets and (B2 only) 261019/screenshots. Part G's julia files stay on the julia PC (data/juliaracer/): nothing to copy. A short text note of anything you noticed helps.
5. **The on-track parts need no new sheets:** they use 261010_rung_official and 261005_jr_ww103 unchanged, and julia has both sheets from 18 October. If you do change a setup for a run, export a sheet of it and say which run.

Part A — Pedal check and full-throttle pulls (15 min, do this first)

A1 Check the accelerator reaches 100 % (3 min). In iRacing: Options → Controls, press the accelerator fully and watch its bar. If it stops short of the end, run iRacing's **Calibrate** wizard for the pedals, pressing each pedal to the floor when asked. Do the same check for the brake and clutch. No screenshot is needed: the A2 telemetry records the pedal.

A2 Full-throttle pulls in 4th, both setups (12 min). Monza main straight: from about 120 km/h in 4th, **floor the accelerator** up to the braking point. Three pulls

with the default setup, three with WW103. Keep the pedal on the floor the whole way.

Why: on 18 October the pulls topped out at 80.7 % pedal. julia matches your 80.7 % pulls to 98 %; a true 100 % pull gives the engine's real top end.

Part B — Garage questions (20 min, no driving)

Base setup 261010_rung_official at Monza GP without chicanes, as on 18 October. Every sheet name below starts with 261019_ so it can't be mixed up with the 18 October sheets.

B1 Tyre pressure range (3 min). Tires tab, left front: click the cold pressure down until it **stops changing** (its minimum), wait, export the sheet; then up until it stops changing (its maximum), wait, export. Reload the base.

Sheet	Left-front cold pressure
261019_B1_pressure_min	minimum
261019_B1_pressure_max	maximum (keep clicking until the number stops moving)

B2 The drop-down lists (4 min, **screenshots**: a sheet shows only the chosen value). Open each list and screenshot it: steering ratio (Chassis, Front), rear ARB diameter (Chassis, Rear), final drive and each of the five gears (Drivetrain).

B3 Bar stiffness per hole, with a twist (13 min). Export a sheet after each step:

1. LF spring perch +**25 clicks** from the base (this twists the car so the front bar shows in the corner weights).
2. Keeping that twist, set the front ARB arms to **soft, 3, 4, 5, 6, firm** in turn (2 is the base, step 1's sheet).
3. Then, arms back at 2, the front diameters **Disconnected** and **0.4375"** (0.375" is the base).
4. Reload the base. LR perch +**25 clicks**; the rear arms **soft, 2, 3, 4, 6, firm** (5 is the base), then, arms back at 5, **every other rear diameter** in the list (0.6875" is the base).

Sheet	Twist	ARB arms	ARB diameter
261019_B3_twistF	LF perch +25	front 2 (base)	front 0.375" (base)
261019_B3_twistF_arms_soft	LF perch	front soft	0.375"

Sheet	Twist	ARB arms	ARB diameter
	+25		
261019_B3_twistF_arms_3	LF perch +25	front 3	0.375”
261019_B3_twistF_arms_4	LF perch +25	front 4	0.375”
261019_B3_twistF_arms_5	LF perch +25	front 5	0.375”
261019_B3_twistF_arms_6	LF perch +25	front 6	0.375”
261019_B3_twistF_arms_firm	LF perch +25	front firm	0.375”
261019_B3_twistF_diam_disconnected	LF perch +25	front 2	front Disconnected
261019_B3_twistF_diam_0p4375	LF perch +25	front 2	front 0.4375”
261019_B3_twistR	LR perch +25	rear 5 (base)	rear 0.6875” (base)
261019_B3_twistR_arms_soft	LR perch +25	rear soft	0.6875”
261019_B3_twistR_arms_2	LR perch +25	rear 2	0.6875”
261019_B3_twistR_arms_3	LR perch +25	rear 3	0.6875”
261019_B3_twistR_arms_4	LR perch +25	rear 4	0.6875”
261019_B3_twistR_arms_6	LR perch +25	rear 6	0.6875”
261019_B3_twistR_arms_firm	LR perch +25	rear firm	0.6875”
261019_B3_twistR_diam_disconnected	LR perch +25	rear 5	rear Disconnected
261019_B3_twistR_diam_<value>	LR perch +25	rear 5	each other rear diameter, written with p

Sheet	Twist	ARB arms	ARB diameter
			for the point (0.625” → 0p625)

Why: the 18 October roll runs measured the bars at soft, 2, firm (front) and soft, 5, firm (rear). The other holes and the other diameters are still interpolated in julia. The corner weights under a twist give each one directly.

Part C — Part-throttle steps on the straight (15 min, default setup)

Monza main straight, 4th gear. From about 120 km/h hold the accelerator **steady at about 25 %** until the braking point. Then the same at **50 %** and **75 %.** Twice each. Exact percentages don’t matter, but **steady** does: rest your foot, don’t chase a speed.

Why: julia’s part-throttle map was fitted before it learned about air density, and at 25–75 % it now differs from your files. Steady pedal positions give the map point by point.

Part D — Skidpad: throttle and brake steps on a steady circle (25 min)

D1 Throttle steps (12 min). Skidpad, default setup, clockwise. Settle on a steady circle at about **0.8 g** (a bit below the limit). Then, **without changing the steering**, add throttle in three steps, about **+10 %, +20 %, +30 %**, holding each for 5 seconds. Let the car run wide or step out, whichever it does; just note which. Back to the steady circle between steps. Three times, then anticlockwise, then WW103 in both directions.

D2 Brake steps (8 min). The same steady 0.8 g circle, default setup. Brake lightly in three steps, about **10 %, 20 %, 30 %** pedal, 3 seconds each, steering unchanged. Twice each direction.

D3 One steady power slide (5 min, optional). As 18 October’s D1: at the limit, add throttle until the rear steps out a little and hold that slide for 10 seconds. Once per direction, default setup. (Your 18 October slides are good; one more pair with the pedal at a true 100 % available is a cross-check.)

Why: julia's car needs 72 % throttle to hold a slide that iRacing holds at 48 %: its rear tyres lose too little grip when they also have to drive the car. Throttle and brake steps on a fixed circle show how much side grip each amount of drive or braking costs, which is the shape julia's tyre model needs.

Part E — Firm trail braking at 250 km/h (15 min)

Monza, the run to Curva Grande, default setup:

1. At about 250 km/h, brake firmly, about **60 % pedal**, in a straight line; release and turn in. Reference, twice.
2. Now **keep about 50 % brake on while turning in**, with more steering than on 18 October: the car should be turning noticeably while still braking hard. Three times. If the rear steps out, that is the result we want; keep it safe.
3. The same at the **Parabolica** entry (from about 230 km/h), three times.
4. Repeat 1–3 with WW103.

Why: on 18 October the trail braking was gentle (about 40 % pedal, little steering), and julia and iRacing differed by less than 2° of slide angle, which doesn't settle anything. Firmer braking into a real turn-in shows whether julia's car is as stable as iRacing's at 250 km/h.

Part F — Controlled car-to-car contacts (25 min)

An AI session as on 18 October (Monza, Lotus 49 AI if offered, lowest AI skill, 3 cars, practice or race, damage on). Do these at **low speed and on a straight**: they are about what contact does, not about racing.

#	Do	Closing speed	Watch for
F1	Your front wheel touches an AI car's **rear wheel** from behind (wheel to wheel, slightly offset)	5, then 15 km/h	Does either car spin? Does one ride up?
F2	Side by side, steer gently into the AI car so **the side**	0–5 km/h sideways	Do both cars turn? Which way?

#	Do	Closing speed	Watch for
	wheels rub		
F3	Your **front wheel against its rear wheel** with more overlap, from behind	20 km/h	Does a car launch or flip?
F4	Your nose into its **rear wheel's side** at an angle (a "T-bone" on the wheel)	15 km/h	Spin rate, damage
F5	Optional, last: a 30–40 km/h version of F3	30–40 km/h	The launch

After each contact: drive one slow lap and write one sentence (who spun, which way, did anything leave the ground, damage, incident count). Keep telemetry running throughout. Restart the session if a car is wrecked.

Why: you asked for julia's collisions to behave like iRacing's. In iRacing a touch of a spinning tyre spins both cars and wheel-over-wheel launches them; julia's cars bounce off each other "like beanbags". Your 18 October races have contacts in them, but at unknown closing speeds and overlaps. These controlled ones give the law.

Part G — WW103 side by side: the same tests in iRacing and julia (40 min)

The new sheet ww103_iracing_setup_sheet.pdf (on the stick, next to this plan) is GPL's WW103 carried to iRacing. It comes out the same as your 261005_jr_ww103, and julia's "WW103 fast loose" is built from that setup. Part G drives the same short tests in both sims, back to back, so you can feel and julia can measure whether the two cars behave the same.

Setup. iRacing: load 261005_jr_ww103 and check it against the sheet (or enter the sheet on 261010_rung_official and save it under that name). julia: launcher → **Car setup: WW103 fast loose**, Garage tab **not** changed, practice mode, no AI. julia

writes its own telemetry file for every run (data/juliaracer/*.ibt), in iRacing's format.

Order. Do each group in iRacing first, then straight away the same group in julia, while the feel is fresh. Same gear, same speeds, same pedal amounts: read them off each sim's speed and pedal display.

#	Where (iRacing / julia)	Do	Watch for
G1	Skidpad / Skidpad (test area)	Steady circles at about 60, 80 and 100 km/h, 2 laps each, clockwise then anticlockwise	Steering needed at each speed; body roll; which end goes first at the top speed
G2	Skidpad / Skidpad	From the steady 80 km/h circle, squeeze the throttle to about 50 % over 2 s and hold. 3 times each way	How soon the rear steps out; can you catch it?
G3	Skidpad / Skidpad	Near the limit on the circle, snap the throttle shut and hold the wheel still 3 s. 3 times each way	Does the nose tuck in? Does the rear come round?
G4	Monza GP without chicanes / Monza	2 laps at a steady 8/10 pace, same gears and braking points (Curva Grande, Lesmo, Parabolica)	Turn-in, mid-corner balance, exits on power
G5	Watkins Glen Classic / Watkins Glen	3 laps at your normal pace (WW103 is a Glen setup; iRacing's Classic is GPL's	Lap time; where the car is loose; anything that feels different

#	Where (iRacing / julia)	Do	Watch for
		layout)	

After each group, one line per sim in the note: *"G2 iRacing: rear out after ~1 s, caught it twice / julia: rear out at once, spun twice*". The feel matters as much as the files: "the same", "julia more loose", "julia lazier on turn-in", and so on.

Why: julia's WW103 has been fitted to your 5 October files, but never driven side by side against iRacing on the same day, the same tests, the same driver. The paired files go through the same tools in both directions, so each difference you feel can be found in the numbers (steer per g, roll per g, the slip angle when the rear lets go) and fixed.

What julia does with the files

Files	Tool (JuliaMotorMTK/tools)	Result
A2 (the pedal check rides along)	wot_density_261018.jl, irgold_replay.jl	full power to 9,000 rpm (ENGINERHO-1)
B1 sheets	sheet_frames.py → garage knob table (garage_lotus49.toml)	the pressure range
B2 screenshots	garage knob table	every list knob's values
B3 sheets	sheet_frames.py → garage_fit_261017.jl, arbarms_261018.jl	each bar's stiffness per hole and diameter (RARBARMS-1)
C	irtest_draft_261015.jl (pedal-binned), wot_density_261018.jl	the part-throttle map (ENGINERHO-1 S2)
D1–D3	powerslide_261018.jl, irgold_steady.jl	the rear's grip under drive and braking (UNDERSTEER-1)
E	trailbrake_261018.jl, irgold_replay.jl (inject)	braking stability at speed (TRAILBRAKE-DRIFT)
F	irtest_contact_261015.jl + your notes	the contact law: spin, ride-up, launch (CARCOLLIDE-2)

Files	Tool (JuliaMotorMTK/tools)	Result
G (iRacing + julia pairs)	irgold_steady.jl, powerslide_261018.jl, irgold_replay.jl on both files + your notes	WW103 julia vs iRacing, per test: steer, roll, slide (PHYSFID-1)